Data Creators › Day 3 — run sheet

Day 3 run sheet — dbt practical and consolidation

Instructor-only companion to the anchor notebook, in the notebook's order: the same numbered sections and labelled parts, each with its time, purpose, what is on screen, what to say, what to ask at the whiteboard (with the answers rooms give and what to add), the demo steps with their coaching, fallbacks, the transition and what to do if ahead or behind. Keep the notebook on the projector and this beside it. Times follow the instructor plan; adapt them to the room and keep the sequence.

Before the day

- Section 1 builds a small dbt project from two CSVs in `demos/price_sheet/`, one cell at a time; its first cell resets the folder. Run the section once before the room arrives.
- The walkthrough cells run the same project as the learners' challenge, on DuckDB in `demos/portfolio_analytics/`. Run `bash demos/portfolio_analytics/demo.sh stretch` once before the day; the cells then rebuild pieces on demand.

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Sections

#	Section	Time	What happens
1	dbt from zero: one transformation through the whole life cycle	09:00–09:50	Recap: yesterday's platform in two pictures · Demo: what dbt is for, why use it, then the build: project, profile, seeds, source, model, Jinja, ref, tests, build, docs
2	The dbt practical: launch, Phase 1 and the walkthrough	09:50–13:00	Launch: the pipeline · Challenge 9: dbt practical · Walkthrough: Phase 1 · Break · Walkthrough: Phase 2 · Circulating: Phase 2 to lunch · Lunch
3	Advanced dbt, then complete and extend the practical	13:00–14:20	Demo: a loop, a macro, a custom test, snapshot, incremental, OPTIMIZE and ZORDER, unit test · Challenge 9 continued: Phase 2 and the stretch · Break
4	Consolidation	14:20–15:00	Activity 9b: second look at yesterday's AI-generated dbt · Debrief: what did we learn? · Activity 9: test review in pairs · Close

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Timeline

§	Time	Block	Type	On screen
1	09:00–09:10	Ice-breaker, then recap: yesterday's platform in two pictures	Recap	Anchor › section 1, Ice-breaker box, then Recap (two Day 2 diagrams)
1	09:10–09:50	dbt from zero: one transformation through the whole life cycle	Demo	Anchor › section 1, top to bottom (demos/price_sheet)
2	09:50–10:00	Challenge 9 — dbt practical: launch and first sections	Practical	Anchor › Launch box; learners in challenges/v1/02-dbt-pipeline
2	10:00–10:15	Break	Break	
2	10:15–11:00	Continue dbt practical	Practical	Anchor › Walkthrough cells (for support)
2	11:00–12:00	Continue dbt practical: Phase 2 to lunch	Practical	Anchor › Circulation checklist and symptom table; learners in challenges/v1/02-dbt-pipeline
2	12:00–13:00	Lunch	Break	
3	13:00–13:25	Advanced dbt: a loop, a macro, a custom test, snapshot, incremental, unit test	Demo	Anchor › section 3 demo cells (demos/price_sheet, then demos/portfolio_analytics)

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§	Time	Block	Type	On screen
3	13:25–14:05	Complete and extend dbt practical	Practical	Anchor › Launch box and walkthrough cells; learners in challenges/v1/02-dbt-pipeline
3	14:05–14:20	Break	Break	
4	14:20–14:35	Second look at yesterday's AI-generated dbt	Pair activity	Anchor › Activity 9b card; learners in the project Copilot and Catalyst generated on Day 2
4	14:35–14:45	Consolidation	Debrief	Anchor › Recap table
4	14:45–14:55	Test review in pairs	Pair activity	Anchor › Activity 9 card
4	14:55–15:00	Closing discussion	Debrief	Anchor › Before learners leave

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1. dbt from zero: one transformation through the whole life cycle

Recap — yesterday's platform in two pictures · 09:00–09:10

Purpose. Put everyone on the same page before dbt: where the tables live and what runs the SQL.

On screen. Anchor › section 1, Ice-breaker box, then Recap (two Day 2 diagrams)

Instructor notes.

- Open with the ice-breaker, the first thing in the notebook, above the agenda, two or three minutes: one thing you always double-check, even though it is almost always fine. One answer each; type a few into the notebook. Its link to the day is tests: a check written down once and run every time. Return to it at Step 5.
- Then about seven minutes, two pictures from Day 2, two or three lines each. Ask one question per picture and let the room answer; do not re-teach.
- End by saying where today sits: dbt holds the transformation rules and sends SQL to the platform on these pictures. The tools-together flow is not repeated; the room has seen it several times.
- Anyone who missed Day 2: point them at the Day 2 anchor's section 3 for the full version.

Say.

- "One line each: what is a lakehouse, and what does Databricks do with your SQL?"
- "dbt holds no data and does no computing. It sends SQL; everything on these pictures is what receives it."

Transition. So what is in a dbt project? Two CSVs and an empty folder.

If ahead or behind. Short of time: the under-the-hood picture only.

Demo — one price sheet, from two raw files to a tested, documented table · 09:10–09:50

Purpose. Show every kind of file in a dbt project being written and run once, on a small example, so the challenge's tasks are recognisable: project and profile, seeds, a source, a model, Jinja, ref, materialisation, tests failing and fixed, build, docs.

On screen. Anchor › section 1, top to bottom (demos/price_sheet)

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Instructor notes.

- Before Step 1, two pictures set up the build: the three layers (staging, intermediate, marts: each one's job, folder and prefix) and the plan (four dashed boxes). The plan picture returns after Steps 2 to 7 as a tracker: yellow is what the step built, blue was built earlier, red has a failing test. Point at it each time so the room knows where it is.
- Present from the notebook; no whiteboard. The flow is: what dbt is for (one picture), why use it (the same transformation as a hand-run script that is silently wrong, then the without-and-with table), the path (eight steps on one picture), then the build. About five minutes for what and why, thirty for the steps, five for docs and the map to the challenge.
- In the build, each cell writes one file (%%writefile) or runs one command. Read the file, ask the room to predict the output, run it.
- Learners do not run the demo cells; their hands-on is the challenge, on Databricks. Say so once before Step 1. The demo runs against a local warehouse file on your laptop so each step answers at once; the notebook does not name it, and there is no need to dwell on it.
- About four minutes a step. Steps 5 and 6 are the point: one test fails on three rows with an empty dollar price, one line in staging fixes it, the build goes green. Never skip them.
- Testing is kept to one red-then-green cycle. Step 7 only adds a test written as a SQL file (no price is zero or negative); it passes. There is no second failure and no deduplication.
- Step 4 has a view-or-table beat: the models block of dbt_project.yml sets it per folder (staging and intermediate views, marts tables, as in the challenge), then config() in one model overrides it and is taken out again. It is challenge S6 in two minutes.
- At Step 3, spend two minutes on the Jinja picture: learners meet source(), ref() and config() in the first hour of the challenge. Loops and macros wait until 13:00.
- Run the section once before the room arrives so dbt's first-run delay is gone; the reset cell empties the folder each time.
- Commands are plain: a %cd cell at the start of each demo sets the working folder once, then cells are just !dbt run, !dbt test and so on, with dbt's full output. If you jump to a cell out of order, run that part's %cd cell first, or dbt runs in the other project.

Say.

- "This script ran without an error and three prices are wrong. The SQL is the easy part; dbt is for everything around the SQL."
- "You write a SELECT. dbt writes the CREATE, works out the order, and checks the result."
- "The run was green and three prices were missing. A successful run says the SQL executed, not that the answer is right."
- "A test is only a query for the rows that break a rule. No rows, it passes."

Demo — dbt from zero, eight steps. A project that starts as two CSVs in `demos/price_sheet/seeds/`; each cell writes one file or runs one command.

1. Read each file aloud before running the command that uses it; ask the room to predict the output.

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2. Steps 5 and 6 are the point of the demo: two tests fail on real rows, one line fixes the model, the build goes green. Do not skip them.
3. Step 3's compiled SQL is the Jinja explanation: curly braces in, plain SQL out. Learners meet source(), ref() and config() within the first hour of the challenge.
4. `demos/price_sheet/README.md` lists every step, file and expected result; `bash reset.sh` starts again.

Transition. The same kinds of file, more of them: the Portfolio Analytics challenge.

If ahead or behind. Behind at 09:40: skip Step 7 (the test of your own) and say it is the pattern of C6. Ahead: open the docs site and click through the lineage graph.

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2. The dbt practical: launch, Phase 1 and the walkthrough

Launch — the pipeline you will build today · 09:50–10:00

Purpose. Launch the assigned dbt project, explain the task sequence and the checks, and get everyone into Phase 1 (sources and staging) before the break.

On screen. Anchor › Launch box; learners in challenges/v1/02-dbt-pipeline

Instructor notes.

- Five-minute launch from the anchor: where to work, the first three commands, what a green section looks like, what reset deletes.
- Then circulate with the three-point checklist. Collect questions in the Q&A table for the recap.
- Break at 10:00; most will be in S1 or S2.

Say.

- “The README teaches dbt as you go: concept, task, hint, check, a variation, then the solution. Attempt before you open the solution.”

Challenge 9 — dbt practical: Phase 1 first · 09:50–10:00

Challenge — dbt practical (challenge 02). `challenges/v1/02-dbt-pipeline/README.md` · about 3 hours across the day.

Say.

- “Attempt each section before you open its solution. The solution is for comparing, not for copying.”
- “You can run the raw dbt commands from the project folder. If dbt says it cannot find your credentials, open a fresh terminal so mise loads the environment, or rerun mise run setup:databricks.”

Walkthrough — Phase 1: sources and staging · 09:50–10:00

Demo — the same project, section by section. The instructor’s copy of the challenge 02 solution in `demos/portfolio_analytics/`: one cell per README section, for “what should I see?”.

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1. Same model names, layers, tags and tests; the Databricks-only post-hook is a no-op macro here.
2. Use a cell when someone asks what they should see; `demos/portfolio_analytics/README.md` maps each cell to its README section.
3. Say once: the SQL is the same; only the target differs.

Transition. Break; carry on after it.

If ahead or behind. Anyone who knows dbt: skip to Phase 2 and the stretch list.

> ☕ Break · 15 min · back at 10:15

Walkthrough — Phase 2: intermediate, marts, tests, docs · 10:15–11:00

Purpose. Phase 1 complete for most; Phase 2 started by some; blockers resolved; questions collected.

On screen. Anchor › Walkthrough cells (for support)

Instructor notes.

- Use the anchor's DuckDB walkthrough cells to show an expected result when someone is stuck; never type in their project.
- Access problems to the platform contact; code problems to the pair next to them first.

Say.

- "Could this join pass its tests and still double the reported value?"

Transition. At 11:00 the practical carries on to lunch.

Circulating — Phase 2 to lunch: three questions per learner, and the symptom table · 11:00–12:00

Purpose. Phase 1 complete for everyone; Phase 2 under way for most; each learner asked the three circulation questions once before lunch.

On screen. Anchor › Circulation checklist and symptom table; learners in challenges/v1/02-dbt-pipeline

Instructor notes.

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- No platform session this week: its content (Airflow at CG, Unity Catalog and the three catalog types, contracts, the vended repository) was covered on Day 2, so the hour goes to the practical.
- Walk the room with the three-point checklist; use the symptom table for the stalls; keep adding to the Q&A table for the recap.
- Platform questions that come up: answer from Day 2's platform section, or log them in the Q&A table for the platform team.

Say.

- "Show me the grain of this model, the key on the right-hand side of the join, and what happens to a row with no match."

Transition. Lunch.

If ahead or behind. Anyone still in Phase 1 at 11:30: pair them with a neighbour who has finished S7.

> 🍽 Lunch · 60 min · back at 13:00

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3. Advanced dbt, then complete and extend the practical

Demo — Jinja that writes SQL: a loop, a macro, then the production pieces · 13:00–13:25

Purpose. Show what Jinja does beyond filling in names, and the pieces a production project adds, so learners recognise them in the vended project and in Friday's capstone.

On screen. Anchor › section 3 demo cells (demos/price_sheet, then demos/portfolio_analytics)

Instructor notes.

- Two cells on this morning's price sheet (a for loop, a macro), then four on the Portfolio Analytics copy (dispatching macro and generic test, snapshot, incremental, unit test).
- Before the unit test there are two pictures for the Databricks post-hook in challenge C9: OPTIMIZE compacts small files into large ones; ZORDER BY puts similar rows in the same file so a filter can skip files. Two minutes; it answers the question learners ask when they reach C9.
- After each cell show the compiled SQL: the lesson is that Jinja runs at compile time and the warehouse sees plain SQL.
- None of this is needed to finish the challenge; say so, so nobody abandons Phase 2 for it.

Say.

- "A macro is a function that returns SQL text. If you have pasted the same expression into three models, it wants to be a macro."
- "A snapshot keeps the history of rows that change; an incremental model avoids rebuilding rows that do not."

Transition. Back to the practical: Phase 2, then your choice of stretch.

If ahead or behind. Short of time: the loop, the macro and the snapshot; point at the other cells.

Challenge 9 continued — Phase 2, then your choice of stretch · 13:25–14:05

Purpose. Complete and explain the work; Phase 2 for those on it; stretch (macros, snapshots, incremental, unit tests) for fast finishers.

On screen. Anchor › Launch box and walkthrough cells; learners in challenges/v1/02-dbt-pipeline

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Instructor notes.

- About three and a half hours of practice across the day in total; nobody has to finish Phase 2.
- Early finishers investigate one planted error in a copied project and show the failure and the repair.

Say.

- “A successful build is not evidence. Show me one value you checked a different way.”

Say.

- “Carry on where you stopped. Phase 2 first; nobody has to finish it.”
- “Finished C10? Ask Copilot to add one of the pieces you just saw to your project: a unit test for stg_holdings is the quickest.”

Transition. Break, then the recap.

> ☕ Break · 15 min · back at 14:20

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4. Consolidation

Activity 9b — second look at yesterday’s AI-generated dbt · 14:20–14:35

Purpose. Learners reopen the dbt project they generated yesterday, name two things in it they could not judge then, assess it now that they know dbt, and prompt Copilot again to fix one thing they caught.

On screen. Anchor › Activity 9b card; learners in the project Copilot and Catalyst generated on Day 2

Instructor notes.

- This closes the loop the room opened yesterday: ‘do we trust it, review all of it, or review some of it?’ The answer the activity gives is an order to review in: sources, staging, joins, tests, then anything nobody asked for.
- Yesterday some projects did not finish generating. Those learners pair with a neighbour whose project did, or review the Day 2 example in courseware/day2-platform/demos/contract-to-code/eod_prices.
- Two minutes to write the two things down, eight to review and re-prompt in pairs, five of shareback: two or three pairs give what they caught, the prompt they wrote and whether the fix held. Things still not judged go in the Q&A table.

Say.

- “Think back to yesterday afternoon. Write down two things in that project you could not judge. Can you judge them now?”
- “Yesterday your prompt was: build me a project. Today it can be: the join in this model drops rows with no FX rate; keep them and add a not_null test. That is the difference knowing dbt makes.”
- “You do not review five hundred files. You review in graph order, and you start with the tests: do they guard the promises in your contract?”

Activity 9b — Second look at yesterday’s AI-generated dbt. Pairs, 15 min; the card is in `activities/`.

Transition. From what you caught to what we learned today.

If ahead or behind. Short of time: pairs check only their two things and the tests, and write the prompt without running it.

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Debrief — what did we learn? · 14:35–14:45

Purpose. Ask the room what it learned and fill gaps from the six topics, all from today. Bitemporal loading is not part of today; if tomorrow's plan needs volunteers for it or for the deployment demo, ask for them here.

On screen. Anchor › Recap table

Instructor notes.

- Ask 'what did we learn?' and fill gaps from the recap table rather than lecturing.
- Only if tomorrow's plan still has them: ask for a volunteer for the bitemporal loading slot and one with Harness access for the deployment demo. Check with tomorrow's instructor first; nothing in the notebook asks for this.

Say.

- "What did we learn today? Call them out and I will write them up."

Ask at the whiteboard — what did we learn? (fill gaps from this list, do not lecture it). Ask in order, write what the room says, add what is missing.

Ask	Expect, then add
Models and layers	A model is a SELECT in a file. Staging cleans one raw table (stg_), intermediate joins and applies rules (int_), marts are what consumers read (fct_, agg_).
Dependencies	source() and ref() build the graph; dbt works out the order, and the same graph is the lineage in the docs.
Tests	A test is a query for the rows that break a rule. Generic tests in YAML, your own as a SQL file; dbt build runs them with the models.
Views and tables	Set per folder in dbt_project.yml; config() in a model overrides it. Incremental and snapshot for large tables and for history.
Jinja and macros	Curly braces are replaced before the warehouse sees the SQL; a macro is a rule written once.

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Ask	Expect, then add
Reviewing generated code	Read it in graph order: sources, staging, joins, tests. Yesterday you could not judge it; today you can.

Transition. Pairs: swap a mapping and its test.

Activity 9 — test review in pairs · 14:45–14:55

Purpose. Pairs take one test, name a wrong result that would get past it, and agree one extra test.

On screen. Anchor › Activity 9 card

Instructor notes.

- If a pair is stuck for a wrong result, offer one: a join that doubles the rows, a missing FX rate, a value in the wrong unit.

Say.

- “What wrong result would still get past this test? What one test would catch it?”

Activity 9 — Test review in pairs. Pairs, 10 min; the card is in `activities/`.

Transition. Two or three examples to the room, then close.

Close — closing discussion · 14:55–15:00

Purpose. Share a missing check; save the dbt work for orchestration; confirm tomorrow’s baseline.

On screen. Anchor › Before learners leave

Instructor notes.

- Completing advanced dbt is not a prerequisite for tomorrow; the Airflow challenge ships the solved dbt project.

Say.

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- “Keep this project where it is. Tomorrow Airflow runs it.”

Say.

- “Two or three examples to the room: the wrong result, whether the test caught it, the improvement.”
- “Save the project where it is; tomorrow Airflow runs it. Completing Phase 2 is not a prerequisite for tomorrow: the Airflow challenge ships the solved dbt project.”

Before learners leave

- [ ] Two things each learner could not judge in yesterday’s generated project, marked as judged or still open; one thing they caught, and the prompt that fixed it.
- [ ] The working dbt project, model and test results, one independent output check and the outstanding questions for orchestration, kept where they are.
- [ ] Each learner can trace one plain-English mapping through its model and its test.
- [ ] The Airflow baseline is ready: Databricks auth works, challenge 02’s seeds are loaded, and Astro has been started once after setup.

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